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A comprehensive large-scale biomedical knowledge graph for AI-powered data-driven biomedical research

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To address the rapid growth of scientific publications and data in biomedical research, knowledge graphs (KGs) have become a critical tool for integrating large volumes of heterogeneous data to enable efficient information retrieval and automated knowledge discovery. However, transforming unstructured scientific literature into KGs remains a significant challenge, with previous methods unable to achieve human-level accuracy. Here we used an information extraction pipeline that won first place in the LitCoin Natural Language Processing Challenge (2022) to construct a large-scale KG named iKraph using all PubMed abstracts. The extracted information matches human expert annotations and significantly exceeds the content of manually curated public databases. To enhance the KG's comprehensiveness, we integrated relation data from 40 public databases and relation information inferred from high-throughput genomics data. This KG facilitates rigorous performance evaluation of automated knowledge discovery, which was infeasible in previous studies. We designed an interpretable, probabilistic-based inference method to identify indirect causal relations and applied it to real-time COVID-19 drug repurposing from March 2020 to May 2023. Our method identified around 1,200 candidate drugs in the first 4 months, with one-third of those discovered in the first 2 months later supported by clinical trials or PubMed publications. These outcomes are very challenging to attain through alternative approaches that lack a thorough understanding of the existing literature. A cloud-based platform (https://biokde.insilicom.com) was developed for academic users to access this rich structured data and associated tools.

The sheer volume of information produced daily in scientific literature, expressed in natural languages, makes it impractical to manually read all publications, even within relatively narrow research areas. Additionally, advances in high-throughput technologies have led to the creation of enormous quantities of research data, much of which remains underused in various databases. This information explosion poses a major challenge for researchers to identify and develop innovative ideas using all the available data. Automated knowledge discovery (AKD; also known as automated hypothesis generation) can help mitigate this problem by automating the process of data analysis, identifying patterns and generating innovative insights and hypotheses[1]. In recent years, knowledge graphs (KGs) have been proposed as a powerful data structure for integrating heterogeneous data and for AKD[2–8]. KGs, with entities as nodes and their relationships as edges, represent human knowledge in a structured form, facilitating efficient and accurate information retrieval. Graph algorithms can be

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used on KGs to infer potential relationships as plausible hypotheses between known entities.

Computational construction of KGs from unstructured text entails two steps: named entity recognition (NER) to identify key biological entities and relation extraction to extract relationships among entities. Historically, NER and relation extraction have been collectively referred to as information retrieval tasks. Early automated methods mainly fell into two categories: rule-based and machine learning-based. The rule-based approach systematically extracted specific data based on predefined rules[9–14], while the machine learning-based approaches inferred rules from annotated data usually with increased recall and overall performance[15–30]. The advent of machine learning led to more sophisticated methods that leveraged semantic information and sentence structure, resulting in substantial improvements in information extraction effectiveness[21,24]. However, a gap remained compared to human proficiency.

The emergence of deep learning models has allowed for a more nuanced use of information, such as semantic content and grammatical structures. By expanding the use of features and enhancing expressive capabilities, deep models have significantly improved the overall effectiveness of information extraction[31–43]. Recently, the technique of pretraining and large language models (LLMs) have garnered considerable attention, expanding both model complexity and the amount of training data and achieving remarkable progress in information retrieval tasks[43–54]. This was evidenced by the results in the BioCreative VII Challenge in 2021, where fine-tuning of BERT-based models was widely used and the top performance in some tasks closely matched human annotator performance. Subsequently, a highly advanced series of pretrained models, such as GPT-4, emerged[55–57]. These models have been proved to perform comparably or better to humans in multiple tasks, marking a significant breakthrough in the field.

Recently, LLMs such as GPT-4 have been explored for their integration with KGs, aiming to enhance tasks such as NER, relation extraction and event detection through techniques such as zero-shot prompting, in-context learning and multi-turn question answering[58–60]. While these models excel in generalization and large-scale data processing, they still struggle with domain-specific challenges, including handling long-tail entities[61], directional entailments[62] and inconsistencies in retrieving knowledge from paraphrased or low-frequency phenomena[63,64]. Experiments[60] on datasets such as DuIE2.0 (ref. 65), Re-TACRED[66] and SciERC[67] highlight that fine-tuned small models continue to outperform GPT-like LLMs in KG-related tasks. Despite these limitations, LLMs have shown remarkable adaptability in augmenting KGs, particularly when structured data are limited, positioning them as a complementary tool for AKD[58].

To facilitate the methodology development and identification of the most effective methods for KG construction, the National Center for Advancing Translational Sciences of the National Institutes of Health (NIH) organized the LitCoin Natural Language Processing (NLP) Challenge between November 2021 and February 2022. In the LitCoin NLP Challenge dataset, six common biological entity types were annotated: diseases, genes and proteins, chemical compounds, species, genetic variants and cell lines. Eight relation types were also annotated for the entities: association, binding, comparison, conversion, cotreatment, drug interaction, positive correlation and negative correlation. These entity types and relations are highly relevant in translational research and drug discoveries. Our team, JZhangLab@FSU, participated in the challenge and won first place (https://ncats.nih.gov/funding/challenges/winners/litcoin-nlp).

In this study, we applied our information retrieval pipeline developed for the LitCoin NLP Challenge to all PubMed abstracts (cut-off date May 2023) to construct a large-scale biomedical KG (iKraph, the abbreviation for Insilicom's Knowledge Graph). Manual verification confirmed the pipeline's accuracy at a human annotator level. By annotating the directions for the relations in the LitCoin dataset and training a model to predict the direction of relations, we constructed a causal KG capable of making indirect causal inferences. To further enhance the coverage of iKraph, we integrated relation data from public databases and high-throughput genomics datasets, making it the most comprehensive, high-quality biomedical KG to date. To make causal inferences among the entities that are not directly connected in the KG, we designed a probabilistic-based approach, probabilistic semantic reasoning (PSR). PSR is highly interpretable as it directly infers indirect relations using direct relations through straightforward reasoning principles.

Navigating the modern drug development terrain is intricate and resource intensive[68]. The ascent in costs largely stems from previous research exhausting more straightforward drug targets, necessitating a shift towards more complex ones[69]. In this setting, KGs play a pivotal role in AKD[2,70–72], particularly in the domain of drug target identification and drug repurposing[73–77]. A significant challenge in developing methods for such applications has been to comprehensively assess the effectiveness of these studies. For example, in the case of drug repurposing, collecting all the known therapeutic associations of a particular disease or drug requires a thorough search of the literature. Without such knowledge, it is impossible to rigorously evaluate drug repurposing methods. In our investigation, for each repurposing objective, we extracted all therapeutic associations documented in PubMed abstracts. This enabled us to measure recall and observed positive rate (OPR), which is infeasible in previous drug repurposing research.

We demonstrate the power of our approach by conducting several drug repurposing studies: drug repurposing for COVID-19, cystic fibrosis, ten diseases without satisfactory treatment and ten commonly prescribed drugs. For COVID-19 and cystic fibrosis, we performed retrospective, real-time drug repurposing exercises. Our method identified numerous viable candidates, supported by substantial literature evidence connecting the drug and disease entities. This level of interpretability is invaluable when determining the necessity of subsequent research endeavours.

Results

Building a large-scale biomedical KG

To facilitate the methodology development and identification of the most effective methods for KG construction, NIH organized the LitCoin NLP Challenge between November 2021 and February 2022 (https://ncats.nih.gov/funding/challenges/litcoin). Our team, JZhangLab@FSU, participated in the challenge and won first place (Table 1). In the summer of 2023, we also participated in the BioRED (Biomedical Relation Extraction Dataset) track of the BioCreative Challenge VIII. In the end-to-end KG construction task, our team also achieved the highest score[78] (Table 1). The LitCoin NLP Challenge dataset comprises 500 PubMed abstracts, each annotated with six distinct entity types and eight types of relation at the abstract level. We used the pipeline initially developed for the LitCoin NLP Challenge to process all PubMed abstracts available before May 2023, creating a large-scale KG, iKraph. In constructing iKraph, we processed more than 34 million PubMed abstracts, resulting in 10,686,927 unique entities and 30,758,640 unique relations. We incorporated entity normalization into our pipeline, as this was not a component of the LitCoin NLP Challenge (see Supplementary Information Sections 1.2 and 1.3 for more details).

We evaluated the accuracy of our large-scale relation extraction and our novelty prediction results using a sample of 50 randomly selected PubMed abstracts, including 1,583 entity pairs. The results shown in Supplementary Table 3 indicate that our information extraction performance rivals that of human annotations. A more in-depth analysis is available in Supplementary Information Section 2.

Figure 1a shows the number of PubMed abstracts containing one or more of the four major types of entity: diseases, genes, chemicals and sequence variants. It is evident that diseases are the most common topic, with more than 20 million articles referencing at least

Table 1 | The performance of the top teams in the LitCoin NLP Challenge and BioCreative Challenge VIII (BC8) BioRED track subtask 2 (end-to-end KG construction)

Competition	Team	NER	RE	Overall	+ID	+Entity pair	+Relation type	+Novelty
LitCoin	JZhangLab@FSU	0.9177	**0.6332**	**0.7186**	–	–	–	–
	UTHealth SBMI	**0.9309**	0.5941	0.6951	–	–	–	–
	UIUCBioNLP	0.9068	0.5681	0.5681	–	–	–	–
BC8	156 (Insilicom)	**0.8926**	–	–	0.8407	**0.5584**	**0.4303**	**0.3275**
	129	0.7858	–	–	0.7635	0.4127	0.3103	0.2334
	127	0.7830	–	–	0.7598	0.3945	0.2976	0.2280
	118	0.7831	–	–	0.7561	0.3903	0.2859	0.2248
	GPT-3.5+PubTator3	0.8652	–	–	**0.8565**	0.3021	0.1081	0.0714
	GPT-4+PubTator3	0.8652	–	–	**0.8565**	0.3449	0.1704	0.1277

The LitCoin NLP Challenge results include NER and relation extraction (RE) scores, reported as Jaccard scores, and the overall score. Our team, JZhangLab@FSU, achieved the highest overall score. The BioCreative VIII BioRED track subtask 2 results present F_1 scores for various evaluation metrics. Our team, Insilicom, ranked first. '+' indicates additional task-specific scores. Bold values indicate the highest-performing scores within each column in each competition. ID is entity normalization, which is the term used in the BioCreative Challenge for ID normalization.

one disease entity, and nearly half of these articles focus exclusively on diseases. In contrast, gene mentions often coexist with other entities, such as chemicals and diseases. Figure 1b depicts the number of PubMed abstracts containing one or more of the five major types of relation, offering insight into the distribution of topics in biomedical research.

Figure 1c compares the relations extracted from PubMed with those from databases and the LitCoin dataset. There is a clear difference between the LitCoin dataset and general PubMed abstracts, as the former contains more relations in each abstract, especially those involving sequence variant entities[79]. This explains the performance difference of our pipeline on these two datasets. Relations from PubMed and public databases are also complimentary.

Figure 1d shows the number of new discoveries for different entity pairs over the year. We have observed a remarkable upswing in disease–gene relations since 2005, which underscores the tangible outcomes of translational initiatives promoted by federal agencies. Furthermore, the increasing number of disease–gene relations signifies an improved understanding of disease mechanisms at the molecular level, thereby bolstering efforts in drug discovery. Of particular note is the rapid escalation of chemical–disease relations in recent years, particularly around 2020, which is anticipated to continue in the foreseeable future.

We plotted $P(k)$ versus k, where k is the degree of an entity in iKraph and $P(k)$ is the probability of an entity having degree k (Fig. 1e). We found that iKraph exhibits a scale-free topology with an alpha parameter value of around 3.0 (more details in Supplementary Information Section 3).

Supplementary Table 4 compares the numbers of relations for five types of entity pair from all the public databases integrated into iKraph, those extracted from PubMed and the numbers extracted if we use a simple co-occurrence rule, which considers two entities having a relation if they co-occur in an abstract. On the one hand, iKraph has significantly more numbers of relations than those from public databases. On the other hand, the numbers of co-occurrences are much larger than relations extracted from PubMed, indicating a substantial noise reduction by explicitly extracting relations from literature compared to retrieval using keywords.

Constructing a causal KG

We developed a model to predict the direction of correlation relations in the LitCoin dataset by determining the source and target entities for each relation. Adding this directional information transformed correlations into potential causal relationships, allowing us to construct a directed KG for knowledge discovery applications.

PSR for inferring indirect relationships

With directional information, we can infer relations between indirectly connected entities using straightforward reasoning. To this end, we designed the PSR algorithm, which is both efficient and interpretable. PSR enables all-against-all drug repurposing for all drugs and diseases with limited computational resources and allows efficient updates of newly inferred relations. For instance, freshly published PubMed articles can be processed daily to extract discoveries and generate hypotheses for timely dissemination. In contrast, most machine learning methods struggle to achieve this level of efficiency and interpretability.

Drug repurposing for COVID-19 using iKraph

Using the PSR algorithm, we conducted a retrospective, real-time drug repurposing study for COVID-19 spanning from March 2020 to May 2023 (Fig. 2). During this period, we consistently discovered repurposed drugs based on the drug targets reported for COVID-19 between March and June 2020. A candidate drug has at least one directed path to COVID-19 through an intermediate gene. We checked whether any repurposed drugs were later validated by either PubMed literature or clinical trials through a monthly assessment. The validation involved scrutinizing whether these repurposed drugs had been subsequently tested in clinical trials documented on ClinicalTrials.gov or had published therapeutic efficacy in COVID-19 patients in PubMed abstracts. Notably, drugs identified in clinical trials may not always translate into effective treatments for COVID-19. Nevertheless, they serve as valuable hypotheses, aligning with the primary objective of our drug repurposing approach. As shown in Fig. 2a, we were able to identify nearly 600 to 1,200 candidate drugs from iKraph using PSR. One-third of the repurposed drugs identified during the initial 2 months were later validated as effective treatments or plausible potential treatments worthy of clinical trials. Even drugs that did not achieve validation status remain viable hypotheses, warranting further investigation, particularly when existing treatments prove less than optimal.

Figure 2b shows a timeline of repurposed drug validation. Notably, there is a surge in validated drugs during the first year, which subsequently shows a month-to-month decline. This trend suggests most repurposed drugs align with practitioners' early assessments. Some drugs were validated only in the second or third year, indicating they were less immediately evident. The number of drugs validated through publications matches those validated via clinical trials. While numerous drug repurposing studies for COVID-19 exist[80–83], as per our understanding, no previous research has as thoroughly validated such a vast quantity of repurposed drugs as we have in this research. These results highlight iKraph's ability to identify promising drug candidates for specific diseases in real time.

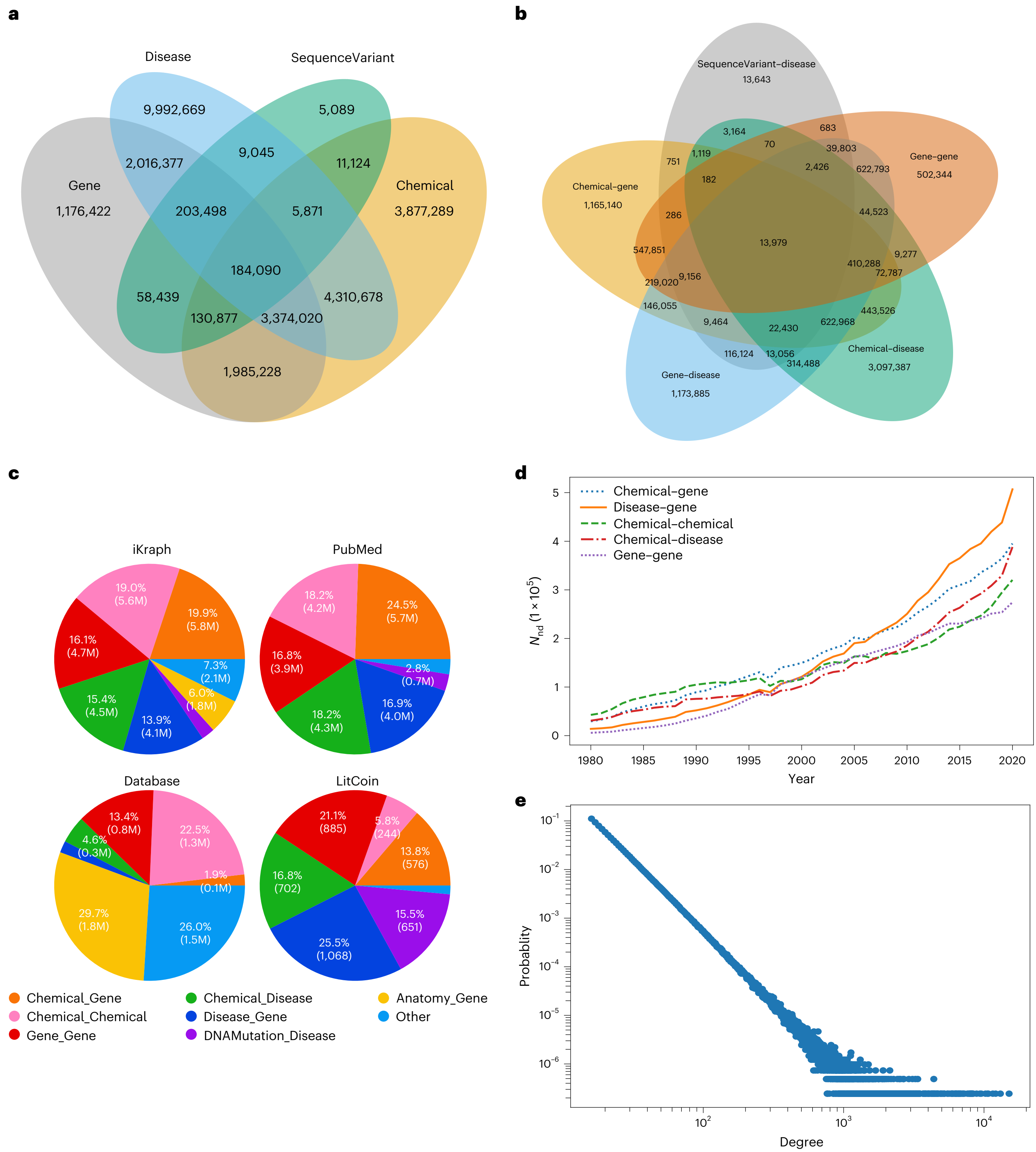

Fig. 1 | The coverage of iKraph and some basic properties. a, Venn diagram for the number of PubMed articles containing certain types of entity. **b**, Venn diagram for the number of PubMed articles containing certain types of relation. **c**, The composition of relations in iKraph, PubMed abstracts, public databases and LitCoin dataset. M, million. **d**, The numbers of new discoveries (N_{nd}) by entity pair type from 1980 to 2020. **e**, Degree distribution of iKraph, where the x axis represents the degree k of an individual entity and the y axis $P(k)$ denotes the corresponding probability of any entity exhibiting that degree.

We then conducted drug repurposing for COVID-19 in the current timeframe (Fig. 2c). We did not exclude drugs already reported as treatments for COVID-19 (direct relations). This was to observe if our repurposing efforts agree with existing treatment choices for COVID-19. Figure 2c displays the top 50 repurposed drugs. Notably, most of these drugs (36 out of 50) have published studies mentioning either their potential therapeutic efficacy or demonstrated therapeutic efficacy for phenotypes associated with COVID-19. Among the remaining 14, 11 have been proposed as potential treatments for COVID-19 (citations provided in Supplementary Table 3). For each drug,

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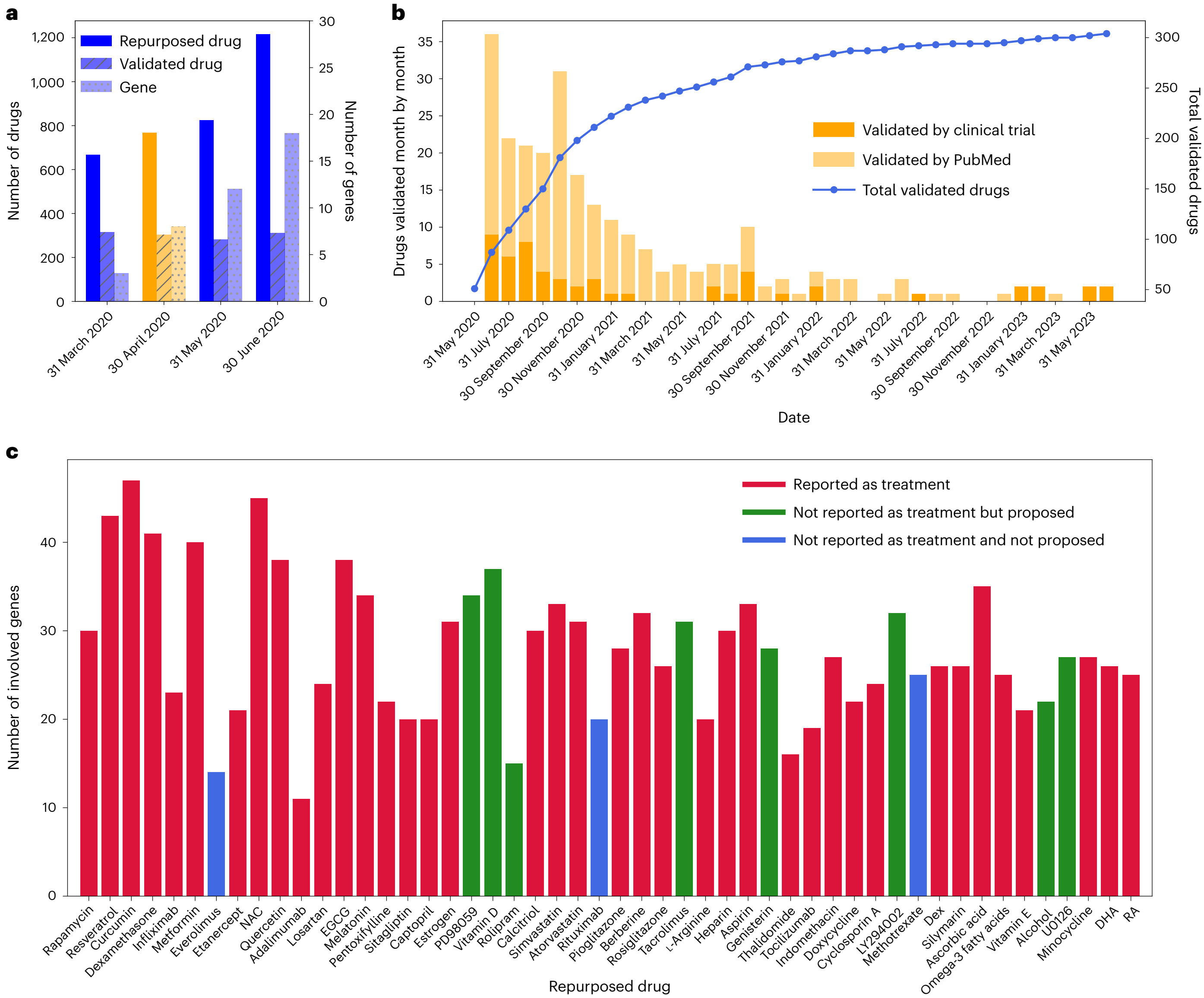

Fig. 2 | Drug repurposing for COVID-19. a, The number of repurposed drugs, number of verified drugs and number of COVID-19-related genes for the first 4 months of the COVID-19 pandemic (March to June 2020). **b**, The number of verified drugs each month for those repurposed for April 2020. **c**, The number of genes involved in the drugs repurposed at present time (March 2023). The figure shows the top 50 repurposed drugs sorted from left to right, with those on the left having higher scores. Almost all the repurposed drugs interact with many genes (height of the bar) related to COVID-19. Most of the drugs were reported as treatment for COVID-19 (36 out of 50). Among those that were not reported as treatments for COVID-19, 11 out of 14 were proposed as potential treatments. NAC, *N*-acetyl cysteine; EGCG; epigallocatechin gallate; DHA, docosahexaenoic acid; RA, retinoic acid.

numerous genes that link COVID-19 with the drug were identified (*y* axis of Fig. 2c). Additionally, each of these relations, whether drug–gene or gene–COVID-19, is supported by one or several articles. To our knowledge, none of the previous literature-based COVID-19 repurposing studies has yielded such comprehensive findings.

Drug repurposing for cystic fibrosis using iKraph

We applied PSR to uncover indirect relationships between drugs and cystic fibrosis from 1985 to 2022 (Fig. 3). Since the early 1990s, at least 50 potential repurposed drugs have been identified annually. A drug was considered validated if later reported as directly therapeutic for cystic fibrosis. Historically, estimating these metrics was challenging due to reliance on manual literature searches. We calculated recall (percentage of known direct relations successfully repurposed) and OPR (percentage of repurposed cases with reported direct relations). Unlike precision, OPR accounts for potential candidates awaiting validation. From 1990 to 2022, the average recall is 0.635 (Fig. 3b), and the average OPR from 1985 to 2011 is 0.159. Different time intervals were used because OPR requires earlier predictions, while recent predictions need time for validation.

We calculated the typical duration for these repurposed drugs to be validated. Our proposed drugs typically appeared in literature 2 to 33 years later, with a median validation time of 9.4 years (Fig. 3a). Assuming experimental validation takes 2 years on average, iKraph could hypothetically reduce this time from 9 to 2 years if predictions were acted on immediately. With over 63% recall and a 9-year median lag, our findings highlight iKraph's potential to accelerate drug repurposing and validation for cystic fibrosis treatment.

Drug repurposing for ten diseases and ten drugs

To evaluate our method's versatility, we extended drug repurposing to ten diseases lacking satisfactory treatments and ten commonly

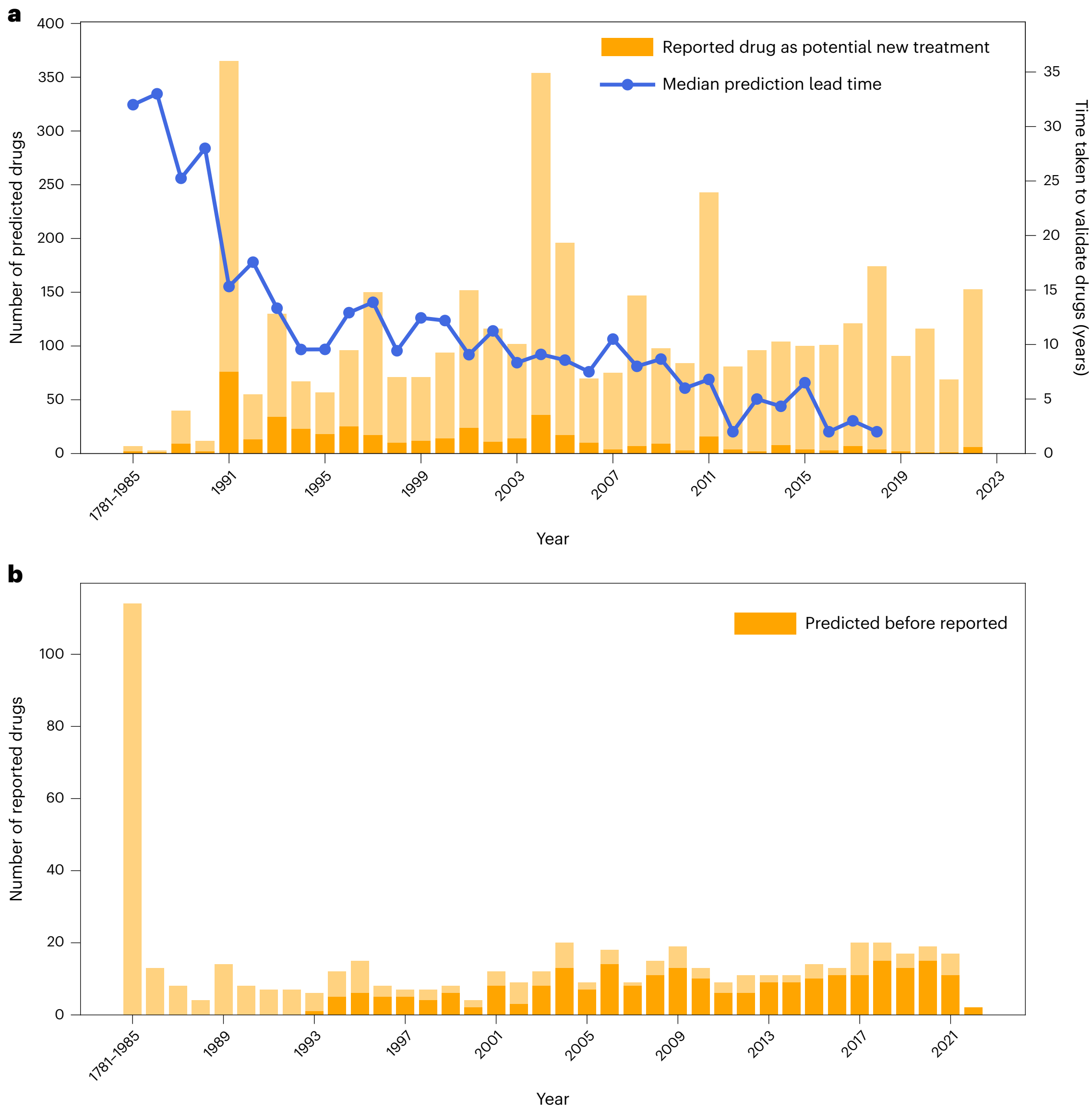

Fig. 3 | Drug repurposing for cystic fibrosis. a, The number of repurposed drugs from 1985 to 2022. The dark yellow bar shows the number of validated drugs. The blue line shows the time it takes to validate the targets for the corresponding years. **b**, The number of drugs reported in PubMed from 1985 to 2022. The dark yellow bar shows the drugs that have been predicted previously.

prescribed drugs (Supplementary Fig. 3). Our PSR algorithm identified a vast array of candidates for these drugs and indications. For each drug (or disease) assessed, we calculated both the recall and the OPR. Our findings revealed average recall values of 0.76 for disease repurposing and 0.86 for drug repurposing. This exceptional recall rate emphasizes the potency of iKraph coupled with our PSR algorithm in spotlighting viable drug repurposing candidates. Notably, these elevated recall rates were achieved without an excessive number of predictions. The observed OPRs remained commendable at 0.197 for diseases and 0.07147 for drugs. A significant proportion of indications repurposed for these drugs are not associated with any treatments in PubMed abstracts. This suggests that certain ailments might still be without treatments, and these widely used drugs could potentially fill those therapeutic gaps.

We extended our analysis by using relations from a database, alongside those extracted from PubMed abstracts, to make drug repurposing predictions. Figure 4 illustrates the F_1 scores for these comparisons based on the top 50 predicted repurposed drugs and the top 250 predicted indications. In each panel, blue bars represent PubMed-based predictions, while orange bars represent database-based predictions. Most repurposed drugs or diseases showed higher F_1 scores using PubMed-derived predictions, probably due to the greater amount of information available in PubMed, which databases cannot match.

Discussion

Converting unstructured scientific literature into structured data has been a long-standing challenge in NLP. Successfully addressing this issue can potentially revolutionize the pace of scientific discoveries. Although numerous studies have been conducted over the years, computational methods have yet to achieve the precision of manual annotation in relation to extraction, posing a significant hurdle. The emergence of LLMs in recent years has ushered in noteworthy advancements in information extraction through LLM fine-tuning. In this paper, we report the use of a human-level information extraction pipeline to construct a large-scale biomedical KG by processing all the abstracts in PubMed. By further integrating relation data from 40 public databases and those analysed from publicly available genomics data, the resulting KG, dubbed iKraph, stands out as perhaps the most all-encompassing

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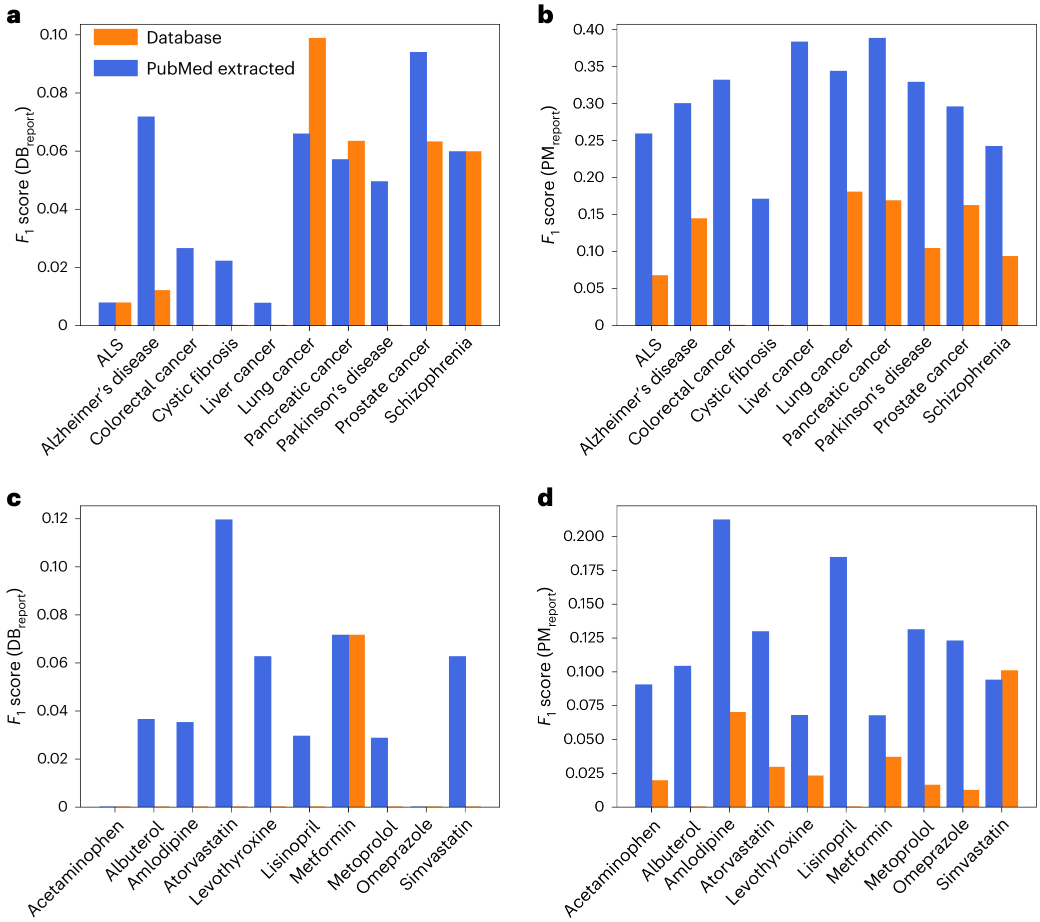

Fig. 4 | F_1 scores for drug repurposing prediction for ten diseases and ten common drugs. The calculation is based on the top 50 repurposed drugs and the top 250 repurposed indications. The blue bars represent predictions made using relations extracted from PubMed (PM) abstracts, and the orange bars represent predictions made using relations from the database (DB). **a**, Validated using therapeutic relations reported in the database for ten diseases. **b**, Validated using therapeutic relations extracted from PubMed abstracts for ten diseases. ALS, amyotrophic lateral sclerosis. **c**, Validated using therapeutic relations reported in the database for ten common drugs. **d**, Validated using therapeutic relations extracted from PubMed abstracts for ten common drugs.

biomedical KG constructed so far. The coverage of iKraph is much larger than public databases for the relations we have extracted. The construction of a causal KG and the design of an interpretable PSR algorithm allows us to perform AKD very effectively. The exhaustive nature of iKraph allows us to perform research that was infeasible previously. We were able to evaluate the performance of AKD systematically and rigorously by calculating recalls and OPRs. Without the knowledge of all PubMed abstracts in a structured form, one must perform a manual search of the literature, which would not be feasible for a relatively large number of predictions. We summarize the notable advances in this study, including some unique iKraph-enabled capabilities in Supplementary Information Box 1 and discuss some of them below.

The biomedical research community has traditionally invested significant resources and human effort in knowledge curation through manual annotations. Our research suggests a paradigm shift, leveraging the capabilities of modern LLMs. By initially producing a limited set of high-calibre labelled data, it is feasible to train an information extraction model that operates at human-level precision on much larger text datasets. This methodology could notably expand the reach of public databases without compromising data quality.

Using iKraph for knowledge discovery tasks, such as drug repurposing, has yielded a vast array of credible candidates supported by an unparalleled volume of literature evidence. This underscores the potential of structured knowledge in hastening scientific breakthroughs. In our drug repurposing endeavours for COVID-19, we highlighted iKraph's proficiency in identifying treatments for pandemics, marking it as an indispensable asset for potential future outbreaks.

Many users might enquire about how iKraph handles noisy information from low-quality papers. Our approach involves aggregating the probabilities of relations (for example, between A and B) across multiple papers (Fig. 5). Each paper assigns a probability to a specific relation, and these probabilities are combined to form an overall score. The more papers that mention the relation, the higher the final probability, making it less susceptible to noise. Relations with low-quality evidence tend to appear in fewer papers, resulting in lower scores. However, while this method provides a strong foundation for handling noisy data, future improvements could involve weighting papers based on factors such as journal impact factor, citation count and publication date. Integrating such metrics aligns with approaches demonstrated in previous work, where features like author diversity, institutional independence, and publication density were found to predict the robustness and reproducibility of scientific claims[84]. Integrating these metrics would allow us to refine the score further by giving more weight to higher-quality sources. For example, papers published in high-impact journals or those widely cited in the scientific community may provide stronger evidence for a relation than those from less reputable sources. Additionally, the publication date can be factored in to balance the relevance of older versus newer findings, ensuring that the most current and impactful research plays a more prominent role in shaping the final probability. This reflects insights from robust scientific

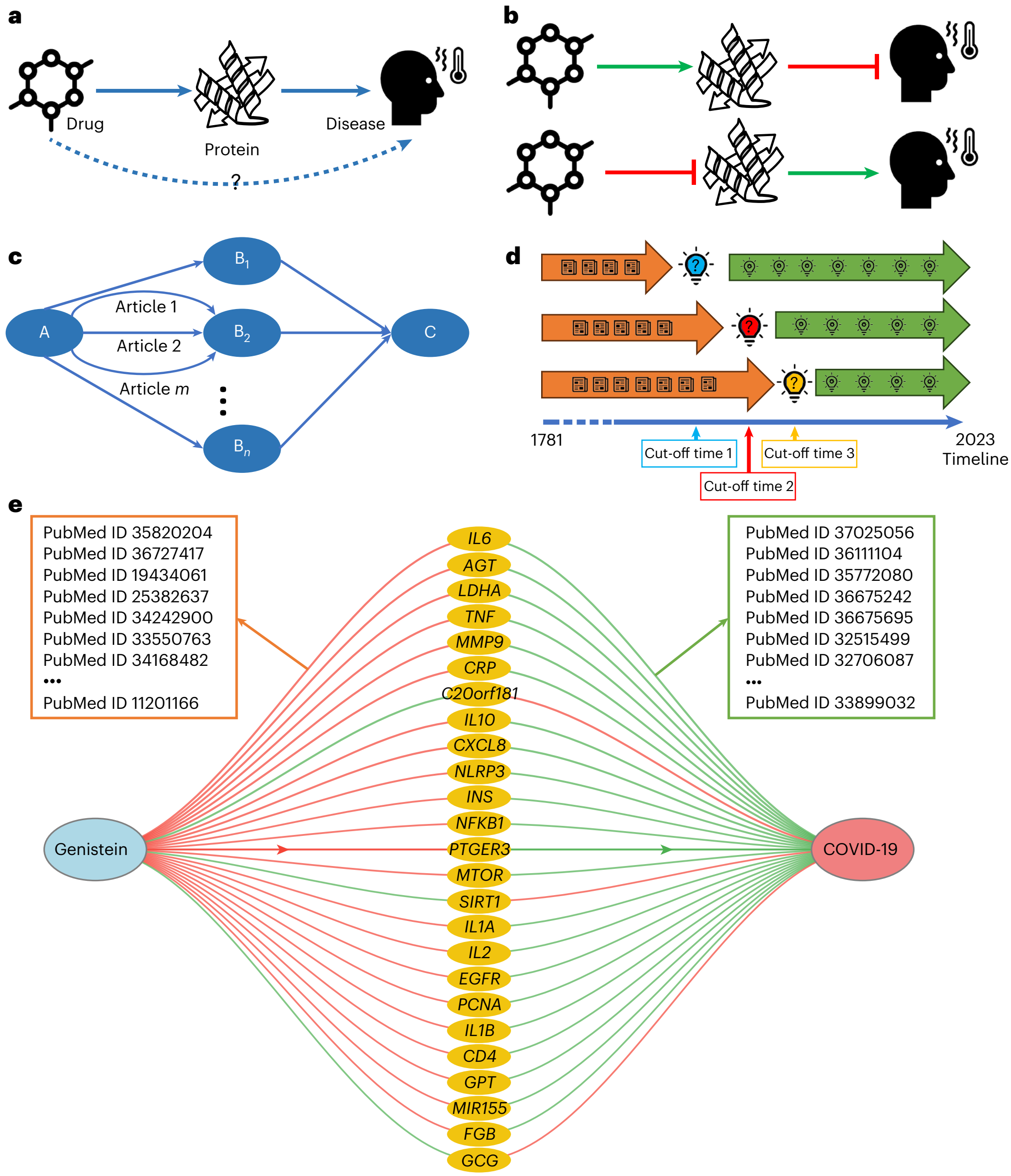

Fig. 5 | The overview of our drug repurposing strategy and validation approach. a, Our method infers drug–disease therapeutic relations through identifying an intermediate entity, the drug target of the disease, with causal relations from the drug to a drug target and from the drug target to the disease. **b**, Two scenarios correspond to a drug–disease therapeutic relation: drug activates a target and the target represses the disease, or drug inhibits the target and the target promotes the disease. **c**, To infer an indirect relation from A (that is a drug) to C (that is a disease), there can be many intermediate potential targets between them. For each relation formed by two entities, there could be many scientific articles mentioning it. The PSR algorithm aggregates all the information to make the inference, while identifying polypharmacological candidates. **d**, To validate a drug repurposing study, we use a time-sensitive approach. We select many cut-off time points (lightbulb with a question mark) and use the knowledge published before the cut-off time (indicated by an orange arrow) to generate predictions and use the knowledge published after the cut-off time (lightbulbs on the green arrow) to validate our predictions. The figure shows three sets of discoveries (blue, red and yellow bulbs) using three different cut-off times. **e**, Genistein's repurposing for COVID-19 treatment. Connected via 25 human genes represented by yellow ovals. The diagram shows drug–gene and gene–disease correlations, with red lines indicating negative correlations and green lines positive ones. Each connecting line is supported by multiple articles from which the correlation probabilities are derived.

literature, where combining high-throughput experimental data with features predictive of reliability has shown promise in assessing the reproducibility of claims[84]. This holistic approach would help iKraph remain robust against misinformation while continuously improving the accuracy of its predictions through adaptive weighting.

Finally, we put our study in the context of the LLMs popular in the current NLP research community. While LLMs have showcased exceptional capabilities in understanding and generating natural language, they are not without shortcomings. A notable limitation is their fixed knowledge cut-off date, which restricts their awareness of the very latest developments. Furthermore, in biomedical research, where precision is crucial, relying solely on LLMs to answer specific questions risks inaccuracies due to their limited knowledge base. Additionally, LLMs possess a propensity to generate text that, while convincingly articulated, may lack factual accuracy. This propensity raises concerns regarding the veracity of answers generated by LLMs, necessitating mechanisms for verification and the production of more substantiated results, possibly with appropriate citations. We believe that integrating KGs such as iKraph with LLMs can effectively mitigate these limitations. To this end, we are actively developing a comprehensive question-answering system, combining iKraph with an open-source LLM.

In Supplementary Information Section 6, we delve into future research avenues and the challenges we have faced. In summary, iKraph serves as a powerful enabler for more effective and efficient information retrieval and AKD.

Methods

Information extraction pipeline

We used the pipeline crafted during the LitCoin NLP Challenge (https://ncats.nih.gov/funding/challenges/litcoin) to process all PubMed abstracts available until 2022, along with data from several renowned biomedical databases, leading to the creation of the KG, iKraph. The construction of iKraph involves three primary stages: NER, relation extraction and novelty classification. The details of the methods can be found in the Supplementary Methods. When developing the pipeline for LitCoin Challenge, we tested a large set of pretrained language models including BERT[46], BioBERT[48], PubMedBERT[85] abstract only, PubMedBERT fulltext, sentence BERT[86], RoBERTa[87], T5 (ref. 88), BlueBERT[89], SciBERT[54] and ClinicalBERT[90]. We tested many ideas, such as different loss functions, data augmentations, different settings of label smoothing, different ways of ensemble learning and so on. Our final pipelines contain the following components: (1) improved in-house script for data processing, including sentence split, tokenization and entity tagging; (2) RoBERTa large and PubMedBERT models as baseline models for NER task; (3) ensemble modelling strategy that combines models trained with different parameter settings, different random seeds and at different epochs for both NER and relation extraction; (4) label smoothing for both NER and relation extraction; (5) using Ab3P (ref. 91) for handling abbreviations for NER; (6) modified classification rule tailored for LitCoin scoring method and (7) training a multi-sentence model for predicting relations at document level, which gave a very competitive baseline for relation extraction.

We used the pipeline developed in the LitCoin Challenge to process all the abstracts in the PubMed database, which contains more than 34 million abstracts, resulting in 10,686,927 unique entities and 30,758,640 unique relations.

Constructing a causal KG

To infer causal relations, we first annotated causal direction for 4,572 relations in the LitCoin dataset. Among them, 2,009 cases have direction from the first entity to the second, 1,611 cases have direction from the second entity to the first and 952 cases have no direction. This annotation allowed us to train a model for predicting the directions for relations, which achieved an F_1 score of 0.924 in a fivefold cross-validation test on the LitCoin dataset. Using a causal KG, we can infer indirect causal relations more effectively for entities not directly connected in the KG: an essential task in AKD.

To make inferences using the causal KG, we designed a PSR algorithm, which calculates the probability of a true relation between two entities connected directly or indirectly. For two entities with a direct edge (a relation mentioned in the literature), there can be multiple mentions in different articles. It is necessary to estimate an overall probability for this pair, which will be used for estimating probabilities for indirectly connected entity pairs. PSR is highly interpretable, which is key for the validation of predictions.

The overall drug repurposing strategy and validation approach are depicted in Fig. 5, with some details provided in the figure legend.

PSR. To make inferences using the causal KG, we designed a probabilistic framework, PSR, for inferring indirect causal relations. PSR is highly interpretable, which is critical for the validation of predictions. There can be multiple mentions in different articles of two entities with a direct edge (a relation mentioned in the literature). It is necessary to estimate an overall probability for this pair, which will be used for estimating probabilities for indirectly connected entity pairs.

To simplify the discussion, let us assume we want to infer the indirect relation from A to C using direct relations from A to B and the relation from B to C. To infer the indirect relation, we first extract the two direct relations. As mentioned earlier, relation A to B and B to C will probably occur many times in different PubMed abstracts. We calculate the overall probability of whether two entities have a particular relation using the formula

$$P_{\mathrm{A}\to\mathrm{B}} = 1 - \prod_{i=1}^{N}\left(1 - p^{i}_{\mathrm{A}\to\mathrm{B}}\right) \tag{1}$$

In equation (1), $P_{\mathrm{A}\to\mathrm{B}}$ is the overall probability of A–B entity pair having a particular relation, $p^{i}_{\mathrm{A}\to\mathrm{B}}$ is the probability of being true for the ith occurrence of these two entities in a PubMed abstract, $1 - p^{i}_{\mathrm{A}\to\mathrm{B}}$ is the probability of this occurrence being false and $\prod_{i=1}^{N}(1-p^{i}_{\mathrm{A}\to\mathrm{B}})$ is the probability of all the occurrences being false (assuming the predictions for these occurrences are independent). The probability of all occurrences being false, when subtracted by 1, gives the probability that at least one of them is true, which is the desired probability. It is also possible that several different relation types will be inferred for a single pair of entities. Often, only one relation type is true, and others may be wrong predictions. To simplify the inference, we selected the relation type with the highest probability as the true relation type for any pair of entities. In reality, there can be multiple entities linking A to C. We denote one of them as B_j. Then, the probability of A to C through B_j can be calculated as

$$P_{\mathrm{A},\mathrm{B}_j,\mathrm{C}} = P_{\mathrm{A},\mathrm{B}_j} \times P_{\mathrm{B}_j,\mathrm{C}} \tag{2}$$

Equation (2) is straightforward since for the indirect relation between A and C (direction from A to C) to be true, both the direct relations must be true. Again, we assume the predictions for the two direct relations are independent. The probability between A to C through m intermediate nodes can then be calculated as

$$P_{\mathrm{A},\cdot,\mathrm{C}} = 1 - \prod_{i=1}^{m}(1 - P_{\mathrm{A},\mathrm{B}_i,\mathrm{C}}) \tag{3}$$

In equation (3), we use $P_{\mathrm{A},\cdot,\mathrm{C}}$ to denote the probability of the indirect relation between A and C through any intermediate entity and there are m such intermediate entities that link A and C. The argument for this formula is similar to equation (1). Putting equations (1)–(3) together, we get

$$P_{\mathrm{A},\cdot,\mathrm{C}} = 1 - \prod_{i=1}^{m}\left[1 - \left[1 - \prod_{j=1}^{n}\left(1 - p^{j}_{\mathrm{A},\mathrm{B}_i}\right)\right]\cdot\left[1 - \prod_{k=1}^{l}\left(1 - p^{k}_{\mathrm{B}_i,\mathrm{C}}\right)\right]\right] \tag{4}$$

In equation (4), m is the number of entities linking A and C, n is the number of instances of A–B_i relations in literature and l is the number of instances of B_i–C relations in literature. Extending this to multiple intermediate nodes between A and C is relatively straightforward. The above probabilistic framework will allow us to rank all the indirect relations that can be inferred. To infer the relation type (positive correlated or negative correlated) between two entities, which multiple intermediate entities could link, we use 1 to represent positive correlations, −1 to represent negative correlations and 0 to represent unknown correlation type between any two entities connected by a direct edge and multiply all the correlations together. The resulting value, 1, −1 or 0 will give us the correlation type between the two entities. If there is at least one unknown correlation type (0) between the two entities, the overall correlation type is unknown. If there is no 0 and an even number of negative correlations, then the overall correlation type will be a positive correlation; otherwise, it is a negative one. For A–C entity pair to have a non-zero probability, there must be a path from A to C with all the directions going from A to C, such as A → B → D → C, while A → B ← D → C is not a valid path from A to C.

In this paper, we show one application of PSR using our iKraph that calculates the indirect relationship between two entities: discover the repurposed drug. We present two study cases for identifying repurposed drugs for COVID-19 and cystic fibrosis, along with an additional study that involves predicting both repurposed drug candidates for ten common diseases and potential additional uses for ten common drugs. The details can be found in Supplementary Information Section 1.7. Figure 5e illustrates Genistein's repurposing for COVID-19 treatment, interacting with 25 human genes. It negatively affects three genes that have a positive impact on COVID-19 while positively influencing the remaining 22 genes, which are negatively associated with the disease. This evidence supports Genistein's potential as a COVID-19 treatment candidate.

Integrating relations from public databases

To integrate the relations in the public databases, we downloaded the relations from two databases that have integrated data from a large number of databases recently, Hetionet[73] and primeKG[92], where Hetionet has integrated data from 29 databases and primeKG has integrated data from 20 databases. The total number of unique databases from both sources is 38. In addition, we extracted drug–target relations from the Therapeutic Target Database[93] and Gene Ontology annotation[94,95]. In total, we integrated relation data from 40 public databases. The KG covers 12 common entity types: diseases, chemical compounds, species, genes and proteins, mutations, cell lines, anatomy, biological processes, cellular components, molecular function, pathway and pharmacologic class. It covers 53 different relation types. Among them, eight were annotated in the LitCoin dataset: association, positive correlation, negative correlation, bind, cotreatment, comparison, drug interaction and conversion. Other relation types came from public databases. When incorporating relations from public databases to maintain the quality of the resulting KG, we excluded relations generated by high-throughput experiments, which are well-known to have a high proportion of false positives, and those predicted by previous machine learning models.

Incorporating relations from analysing RNA sequencing data

We downloaded more than 300,000 human RNA sequencing profiles from the recount3 database[96]. We performed two types of analysis: differential gene expression analysis and gene regulatory network inference. Differential gene expression analysis gave 92,628 differentially expressed genes for 36 different diseases, which correspond to 92,628 disease–gene relations, either positive or negative correlation, depending on whether the genes are up or downregulated in the corresponding diseases. Gene regulatory network inference gave 101,392 gene regulatory relations overall. We added close to 200,000 additional relations by analysing this RNA sequencing dataset.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets used in this study are available on the GitHub repository at https://github.com/myinsilicom/iKraph (ref. 97). Due to size limitations, additional large datasets can be accessed via Zenodo at https://doi.org/10.5281/ZENODO.14846820 (ref. 98). We used the BioRED dataset to train our NER and relation extraction models, and the BioRED dataset can be accessed through https://ftp.ncbi.nlm.nih.gov/pub/lu/BioRED/. The complete KG is hosted on the cloud-based platform: https://www.biokde.com. The downloadable version of the complete iKraph can be accessed via Zenodo at https://doi.org/10.5281/ZENODO.14846820 (ref. 98). Source data are provided with this paper.

Code availability

The code and datasets generated during this study can be found via the GitHub repository at https://github.com/myinsilicom/iKraph (ref. 97).

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Author contributions
Y.Z., X.S., F.P., K.L., S.T., A.E., Q.H., W.W., Jianan Wang and Jian Wang collected data and developed models and pipelines. Y.Z., F.P. and J. Zhang analysed the data and developed methods. D.S., H.C., J. Zhou, E.Z., B.L., T.Z. and J. Zhang. developed the iExplore platform interface. K.Y. and J. Zhang conceptualized and designed the study. Y.Z., F.P., K.Y. and J. Zhang wrote the paper. X.Q., T.Z. and P.Z. provided consultation and paper revision. J. Zhang supervised the study and is the corresponding author.

Competing interests
J. Zhang and T.Z. are owners of Insilicom LLC. The other authors declare no competing interests.

Additional information
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☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	We wrote our own Python code for data collection from PubMed.
Data analysis	We wrote our own code for data analysis. Python=3.9.18; Ab3P (commit 41130cd) – Available at https://github.com/ncbi-nlp/Ab3P; Other tools and libraries used: Spacy (3.6.1), Scispacy (0.5.3), Scikit-learn (1.3.2), Transformers (4.30.2), NLTK (3.8.1), Matplotlib (3.8.4), Pandas (2.0.3), Numpy (1.26.1), Scipy (1.10.1), Huggingface Hub (0.18.0), Requests (2.31.0), and other dependencies. GitHub Repository for Custom Code: https://github.com/myinsilicom/iKraph.

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Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
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We used BioRED dataset to train our NER and RE model and the dataset can be accessed through: https://ftp.ncbi.nlm.nih.gov/pub/lu/BioRED/
The datasets used in this study are available on the GitHub repository at https://github.com/myinsilicom/iKraph99. Due to size limitations, additional large datasets can be accessed via Google Drive: https://drive.google.com/file/d/1OlIiLj7OZ2M6f65Ws5ZSlU-tXdEQElG9e/view?usp=sharing.
The complete knowledge graph is hosted on the cloud-based platform: https://www.biokde.com. The downloadable version of the complete iKraph can be accessed via Google Drive: https://drive.google.com/file/d/1ImT6wCM3woA9u2_vqmu_Obzglu0eqLi0/view?usp=sharing

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	N/A
Data exclusions	No data were removed
Replication	Due to the nature of AI-driven prediction, repeated runs of the model yield consistent results. Therefore, no additional experimental replications were performed. However, model performance was validated through internal evaluation metrics.
Randomization	N/A, since the study is not experimental
Blinding	N/A, This study is based on AI-driven computational predictions rather than human-involved experiments or manual assessments where blinding would mitigate bias. The model processes data in an automated manner, eliminating the need for blinding.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging